Unit 1

1. Introduction to AI

Module 1.5

1.5 AI Development Frameworks

Tutorial 1.5

1.5 AI Development Frameworks

AI-1.5.1 K1 Identify popular AI development frameworks.

AI Development Frameworks

- An AI development framework is a structured set of tools, libraries, and components that simplify the process of building, training, and deploying AI models
- AI development frameworks support a range of activities, such as:
  - Data preparation
  - Algorithm selection
  - Compilation of models to run on various processors, such as central processing units (CPUs), graphical processing units (GPUs) or Cloud Tensor Processing Units (TPUs)
- The selection of a particular framework depends on aspects such as the programming language used for the implementation and its ease of use
- These development frameworks constantly evolve and sometimes maybe combined or replaced by new frameworks

Some Popular AI Development Frameworks

Apache MxNet

- A deep learning open-source framework used by Amazon for Amazon Web Services (AWS).
- Key Features of Apache MxNet
  - Hybrid Programming Model: Supports both imperative (like PyTorch) and symbolic (like TensorFlow) programming styles
  - Multi-language Support: Compatible with Python, Scala, Julia, R, C++, JavaScript, and more
  - Distributed Training: Efficient scaling across multiple GPUs and machines using Horovod and parameter servers
  - ONNX Compatibility: Allows interoperability with other frameworks via the Open Neural Network Exchange format.
  - Portability: Models can be deployed on mobile, IoT, and serverless environments like AWS Lambda.
- Real-World Applications
    - Computer vision
    - Natural language processing
    - Time series forecasting
    - Cloud-based AI solutions

Some Popular AI Development Frameworks

CNTK

- The Microsoft Cognitive Toolkit (CNTK) is an open-source deep-learning toolkit.
- It was designed to train neural networks efficiently at scale, especially for tasks involving speech, image, and text data.
- Key Features
  - Multi-language Support: Works with Python, C++, C#, and BrainScript.
  - High Performance: Optimized for speed and scalability across multiple GPUs and machines.
  - Flexible Architecture: Supports feedforward, convolutional (CNN), and recurrent (RNN/LSTM) networks.
  - ONNX Compatibility: Models can be exported to the Open Neural Network Exchange format for use in other frameworks.
- Applications
  - Speech Recognition: Used in Microsoft's breakthroughs in conversational AI
  - Image Classification: CNNs for visual tasks
  - Natural Language Processing: Text classification and sentiment analysis
  - Time Series Prediction: Forecasting and anomaly detection
- Current Status
  - CNTK is no longer actively developed but still used in many enterprise and research applications

Some Popular AI Development Frameworks

IBM Watson Studio

- IBM Watson Studio is a powerful platform designed to help data scientists, developers, and analysts build, train, and manage AI and machine learning models across cloud environments.
- Core Capabilities
  - Collaborative Environment: Teams can work together on projects using shared tools and data assets.
  - AutoAI: Automates data preparation, model selection, feature engineering, and hyperparameter tuning.
  - Visual & Code-Based Tools: Supports both drag-and-drop workflows and coding in Python, R, and Scala.
  - Model Deployment: Easily deploy models via REST APIs across any cloud or on-premises infrastructure.
  - AI Governance: Built-in tools for tracking data lineage, model metadata, and ensuring regulatory compliance.
- Use Cases
  - Natural language processing for customer insights
  - Image and video analysis
  - Fraud detection and risk

Some Popular AI Development Frameworks

Keras

- A high-level open-source API, written in the Python language and capable of running on top of TensorFlow and CNTK.
- Key Features
  - User-Friendly: Clean, readable syntax for rapid prototyping and experimentation.
  - Modular Design: Build models using either the Sequential API or the more flexible Functional API.
  - Multi-Backend Support: Initially supported TensorFlow, Theano, and CNTK but now primarily focused on TensorFlow.
  - Cross-Platform: Runs on CPUs, GPUs, TPUs, and supports deployment to mobile and web.
  - Scalable: Can be used for small experiments or large-scale production models.
- Use Cases
  - Image classification and object detection
  - Natural language processing (NLP)
  - Generative models (GANs, VAEs)
  - Reinforcement learnin

Some Popular AI Development Frameworks

PyTorch

- An open-source ML library developed by Meta AI and used for apps applying image processing and natural language processing (NLP).
- Support is provided for both Python and C++ interfaces
- Core Features
  - Dynamic Computation Graphs: Build and modify neural networks on the fly
  - GPU Acceleration: Seamlessly run models on CPUs, GPUs, or even TPUs.
  - TorchScript: Convert models to a graph-based format for optimized deployment.
  - TorchServe: Serve models at scale with REST APIs and monitoring tools.
- Use Cases
  - Computer vision (image classification, object detection)
  - Natural language processing (text generation, sentiment analysis)
  - Reinforcement learning
  - Generative models (GANs, VAEs)
  - Time series forecasting

Some Popular AI Development Frameworks

Scikit-learn

- Scikit-learn is a widely-used open-source Python library for machine learning and data mining.
- It’s built on top of NumPy, SciPy, and Matplotlib, making it highly compatible with the broader Python data science ecosystem.
- Core Features
  - Supervised Learning: Algorithms for classification (e.g., SVM, Random Forest) and regression (e.g., Ridge, Lasso).
  - Unsupervised Learning: Clustering (e.g., K-Means, DBSCAN) and dimensionality reduction (e.g., PCA).
  - Model Selection: Tools for cross-validation, grid search, and performance metrics.
  - Preprocessing: Feature scaling, encoding, and transformation utilities.
  - Pipelines: Streamline workflows by chaining preprocessing and modeling steps.
- Use cases
  - Healthcare: Disease prediction, patient risk stratification
  - Finance: Credit scoring, algorithmic trading, loan default prediction
  - Retail: Customer churn prediction, recommendation systems
  - Manufacturing: Predictive maintenance, defect detection
  - Education: Student performance prediction, adaptive learning systems

Some Popular AI Development Frameworks

TensorFlow

- An open-source ML framework by Google which is based on data flow graphs for scalable machine learning.
- It is used for building and deploying machine learning and deep learning models.
- Core Features
  - Computational Graphs: Models are built as graphs of operations on multidimensional arrays called tensors.
  - Automatic Differentiation: Efficient gradient computation for training neural networks.
  - Scalability: Runs on CPUs, GPUs, TPUs, and across desktops, servers, mobile, and edge devices.
  - Multi-language Support: Primarily Python, but also supports C++, Java, and JavaScript.
  - Modular Ecosystem: Includes tools like TensorFlow Lite, TensorFlow.js, and TensorFlow Extended (TFX).
- Use Cases
  - Computer Vision: Image classification, object detection
  - Natural Language Processing: Text generation, sentiment analysis
  - Reinforcement Learning: Game AI, robotics
  - Time Series Analysis: Forecasting, anomaly detection
  - Generative AI: GANs, VAEs, and diffusion models

THANK YOU!!!